

SOFTWARE ENGINEERING DATA ANALYSIS TECHNIQUES

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ABSTRACT

The purpose of this tutorial is to provide a comprehensive coverage of software engineering data analysis techniques. It will briefly cover the basic product and process metrics, their description, use and interpretation. A systematic approach for analyzing and interpreting software engineering data will be introduced. It explicitly recognizes that metrics data tend to have high dimensionality, are highly correlated and suffer from redundancy. The techniques to be presented become progressively more sophisticated in terms of the underlying theory and analysis as well as in their ability to provide insights into the software project and the development process. The specific techniques to be covered are: statistical analyses; regression modeling; stochastic models; classification trees; and neural networks. Our emphasis is on the so-called data mining techniques within the KDD (knowledge discovery in databases) framework.

Keywords

Software Metrics; Statistical analysis, Regression Modeling; Software Reliability; NHPP; Classification Trees; Neural Networks; Radial Basis Functions; Data Mining; KDD.

INTRODUCTION

In recent years, there has been an increasing emphasis on the use of quantitative measures for monitoring and controlling software projects. As a result, collection of product and process data has become a standard practice in major software development organizations. While there has been a growing emphasis on the collection of metrics data, relatively very little work has been done on the systematic use of appropriate data analysis techniques. For metrics or measurement undertakings to be successful, we feel that it is essential to address both the data collection and analysis activities as conjoined

twins. The proposed tutorial is intended to address this issue by presenting a comprehensive set of analysis techniques for software engineering data.

The current state of practice in the use of metrics data is mostly limited to graphical plots that depict a variety of static distributions and some time-varying plots. Examples of such plots are the histograms of complexity measures, total expenditures as a function of program month, and open and closed fault profiles. While such graphical plots give a simple and useful view of product quality and process status, they are inadequate for providing objective insights. Further, such simplistic analyses are not conducive to comparative assessments, process optimization, or objective predictions.

The purpose of this tutorial is to provide a comprehensive coverage of techniques that either have been found to be useful or are promising candidates for extracting interesting and useful information from software data. Our emphasis is on the so-called data mining techniques within the KDD (knowledge discovery in databases) process. The tutorial will cover both model based (statistical and stochastic) models and data based (classification trees and neural nets) data analysis and modeling techniques. In each case, actual data from large software projects will be used for illustration of analyses and interpretation of results.

DETAILED OUTLINE

The material will be presented within the framework of "metrics-based decision making" (see Figure 1). Details of the topics to be covered are given below. Actual software project data will be used throughout the tutorial to clarify the concepts and to emphasize the utility of the methodology being discussed.

1. DESCRIPTION, USE AND INTERPRETATION OF SOFTWARE METRICS

- (a) Role of metrics and measures in Software Engineering

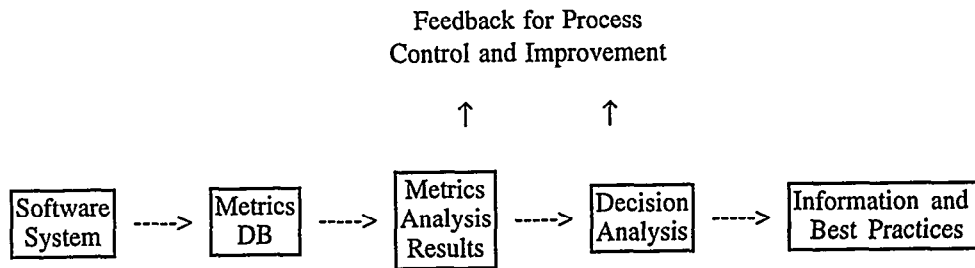


Figure 1. Metrics-Based Decision Making

- (b) Commonly used product and process metrics and their interpretation
 - (c) Brief description of selected measurement programs from industry and government.
2. ANALYSIS, MODELING AND INTERPRETATION USING STATISTICAL METHODS
- (a) Preliminary analysis, data screening and outliers.
 - (b) Basic statistical summaries such as average, variance, and correlation; simple histograms, trend plots and trend tests.
 - (c) Predictive models using regression; basic theory of least squares, stepwise multiple regression; limitations and caution in the use of regression models.
 - (d) Principal component analysis to reduce dimensionality.
 - (e) Principal component regression and its advantages over multiple regression.
3. STOCHASTIC PROCESSES AND RELIABILITY MODELING
- (a) Stochastic processes, Poisson process and non homogeneous Poisson process.
 - (b) Fault profile data analyses using NHPP models.
 - (c) Software readiness assessment based on open and closed faults using a newly developed (ISQ) model.
 - (d) Case studies: commercial and Air Force systems.
4. INTRODUCTION TO CLASSIFICATION TREES AND NEURAL NETWORK MODELS
- (a) Role of CT's and neural models in SE.
 - (b) Basic theory: construction of CT's and neural nets.
 - (c) Use of CT's for early identification of faulty modules, generation of rules.
 - (d) NN's for cost and error estimation.
 - (e) Radial Basis Functions (RBF's): their application and use in SE data analysis.
5. TOOL DESCRIPTION
- (a) SMART: a non-commercial, prototype, SUN-based software data analysis and modeling tool description.
 - (b) Functionality and design features such as information hiding, ease of extension and contraction in functionality, robustness.
 - (c) Demonstration of selected modeling and analysis techniques.

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